

RITESH SAHANI

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PROFESSIONAL SUMMARY

Computer Engineering graduate from TU, IOE Purwanchal Campus with a deep focus on high-quality frontend development and AI/ML integration. Highly proficient in building semantic, accessible, and responsive user interfaces using modern CSS (Flexbox/Grid) and component-based architectures. Experienced in evaluating code quality and reasoning logic, particularly within AI-driven systems. A detail-oriented engineer dedicated to technical accuracy, WCAG standards, and cross-browser compatibility, currently seeking to leverage combined expertise in UI development and AI model training.

TECHNICAL EXPERTISE

- Frontend:** Semantic HTML5, CSS3 (BEM, CSS Modules), Flexbox, CSS Grid
- AI & ML:** LLMs, RAG, Transformers, LangChain, PyTorch, Model Evaluation
- Tools & DevOps:** Git, GitHub, VS Code, Postman, Docker, NPM/Yarn
- Modern Responsive Design, Tailwind CSS, UI:** ReactJS, Cross-Browser Testing
- Core CS:** Data Structures (DSA), DBMS, System Design, Operating Systems
- Languages:** Python, C, C++, JavaScript (ES6+)

SELECTED PROJECTS

RAG-Based Nepali Chatbot & UI Integration

Stack: ReactJS, Python, Django REST, ChromaDB, Hugging Face

- Developed a responsive web interface optimized for multi-script typography (Nepali/English), ensuring high readability and semantic structure.
- Implemented Retrieval-Augmented Generation (RAG) to provide context-aware responses, focusing on technical accuracy and logical consistency.
- Optimized frontend components for efficient data rendering from vector databases with minimal latency.

Real-Time Chat Application Architecture

Stack: MERN, Socket.io, Cloudinary, CSS3 (Custom Styles)

- Engineered a fully responsive UI featuring real-time message updates and state management.

- Applied modern CSS layouts (Flexbox) to ensure seamless cross-device compatibility and intuitive user flow.
- Handled complex UI states including message delivery receipts, typing indicators, and media handling.

Image Colorization Using GANs

Stack: PyTorch, ResNet-18, Matplotlib, NumPy

- Trained deep learning models to predict color values in grayscale imagery using adversarial training.
- Developed a visual interface for model evaluation, demonstrating a methodical approach to ranking AI output quality.

EDUCATION

Bachelor of Engineering in Computer Engineering

2078 – 2082 BS

TU, IOE Purwanchal Campus, Dharan, Nepal

Academic Performance: Consistent honors with up to 76.56% in specialized semesters.

Intermediate (TSLC) & Secondary Education (TSEE)

2074 – 2077 BS

Shree Durga Model Secondary School, Kalaiya

CGPA: 3.44 / 3.56 | Specialized in Technical & Vocational Stream including On-the-Job Training (OJT).

SOFT SKILLS & LANGUAGES

Professional Skills: Technical Writing, Code Review, Accessibility Auditing (WCAG), Systematic Problem Solving, Collaborative Teamwork.

Languages:

English (Professional)

Nepali

Hindi

Bhojpuri